Inviscid Fluid Simulation Through Incompressible Schrödinger Flow Method: A Finite Element Approach

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Introduction

The behaviour of inviscid fluids is described by the Euler equations. These are a set of quasi-linear hyperbolic PDEs, typically expensive to solve due to time step and mesh size requirements [4]. They are usually solved using a Computational Fluid Dynamics approach. This work aims at looking at this problem with a different perspective for predicting the behaviour of inviscid flows: the starting point is the **connection between hydrodynamics and quantum mechanics** proposed by Madelung in 1926 [3]. This idea has been explored by the pioneering work of Chern [1, 2], which developed a novel method under the name of **Incompressible Schrödinger Flow (ISF)** . This work aims at extending the range of applicability of this technique by implementing it using the **Finite Element (FE)** library FEniCSx for Python.

Schrödinger's Smoke

The *Incompressible Schrödinger Flow* is a novel numerical technique, developed in [1, 2], able to **describe inviscid fluids behaviour using the Schrödinger equation** , based on the analogy between hydrodynamics and quantum mechanics [3].

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2}\Delta\Psi + p\Psi \qquad \Longrightarrow \qquad \begin{cases} \nabla\cdot\mathbf{u} = 0 \\ \dfrac{\partial\mathbf{u}}{\partial t} = -\left(\mathbf{u}\cdot\nabla\right)\mathbf{u} - \nabla p \end{cases}$$

One of the most important results of this transformation is the change of a non-linear term $(\mathbf{u}\cdot\nabla)\,\mathbf{u}$ into a linear one $-\frac{\hbar^2}{2}\Delta\Psi$.

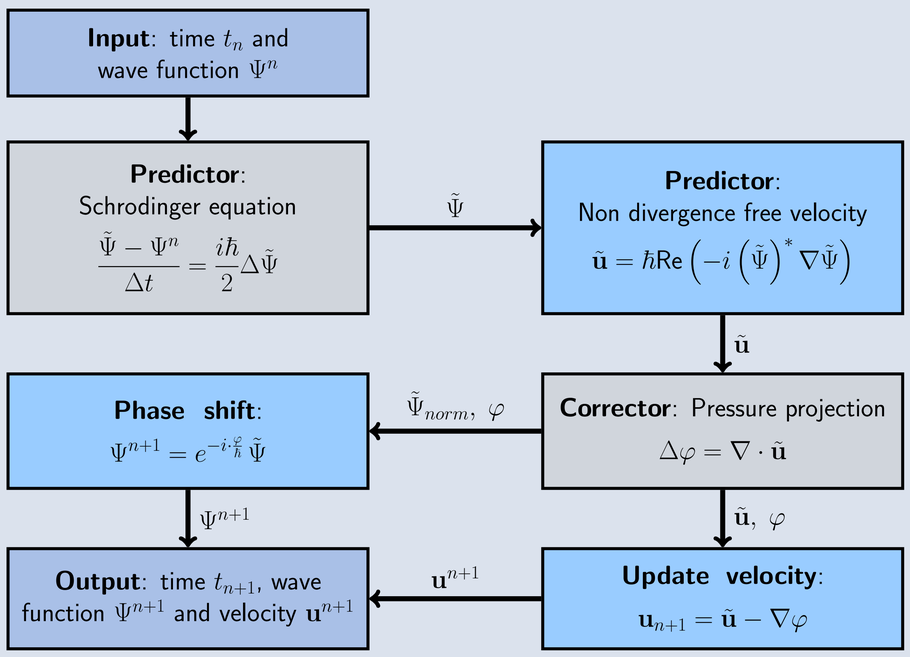

Figure 1. Solution algorithm of ISF.

Numerical Results: Von Karman (MATLAB)

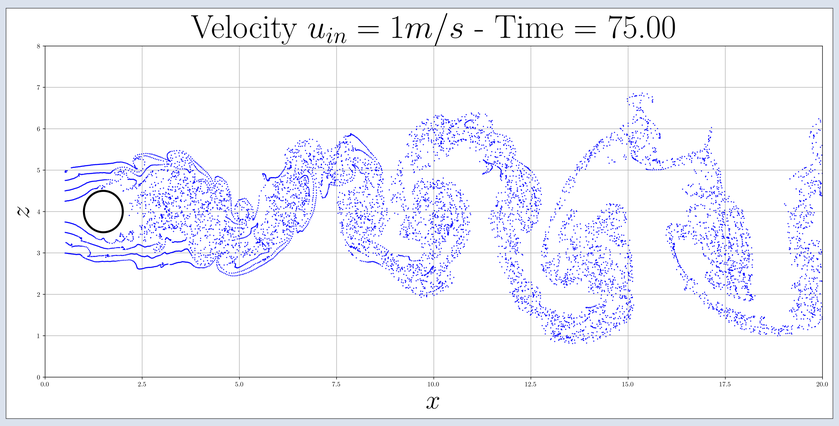

Figure 2. Particles advected by the velocity, computed using ISF (Matlab code, [2]).

- The ISF algorithm is very in **predicting vortex dynamics** , main focus of the original work of [1, 2].
- Using a higher Δt **reduces a lot the solution time** .
- The main drawback of this MATLAB code is that it allows **uniformly structured grids only** and that obstacles are velocity-constrained regions.

References

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[5] H G Weller et al. "A Tensorial Approach to Computational Continuum Mechanics using Object-Oriented Techniques". In: *Computers in Physics* 12.6 (1998), pp. 620–631. DOI: 10.1063/1.168744.

Numerical Results: Backward Facing Step (FEniCSx) - Wave function

- They represent a **plane wave propagation** inside the BFS domain, thus modelling the **pure transport phenomenon** .
- The profiles are very similar to one another, apart from the order of magnitude between ψ_1 and ψ_2, thus the influence of ψ_1 on the velocity is greater than ψ_2.

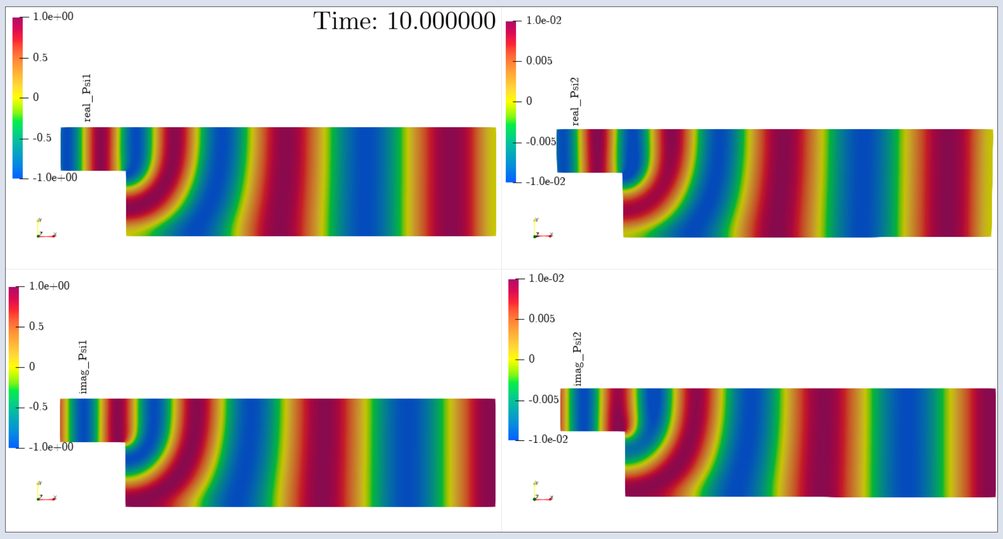

Figure 3. Real and imaginary part of wave function as contour plots (left: ψ_1, right: ψ_2) at final time $T = 10\,s$, using ISF on FEniCSx.

Numerical Results: Backward Facing Step (FEniCSx) - Velocity Profile

- There is **no re-circulation region** behind the step: there is no viscosity, and the flow follows the "wall boundary", keeping the non-penetration condition.
- For the time interval considered, the flow is almost steady.
- The flow is very similar to the potential one, showing that the importance of the vorticity is relatively low.
- The ISF is able to go **beyond potential flows** due to the definition of the $2-$component wave function $\Psi \in \mathbb{C}^2$.

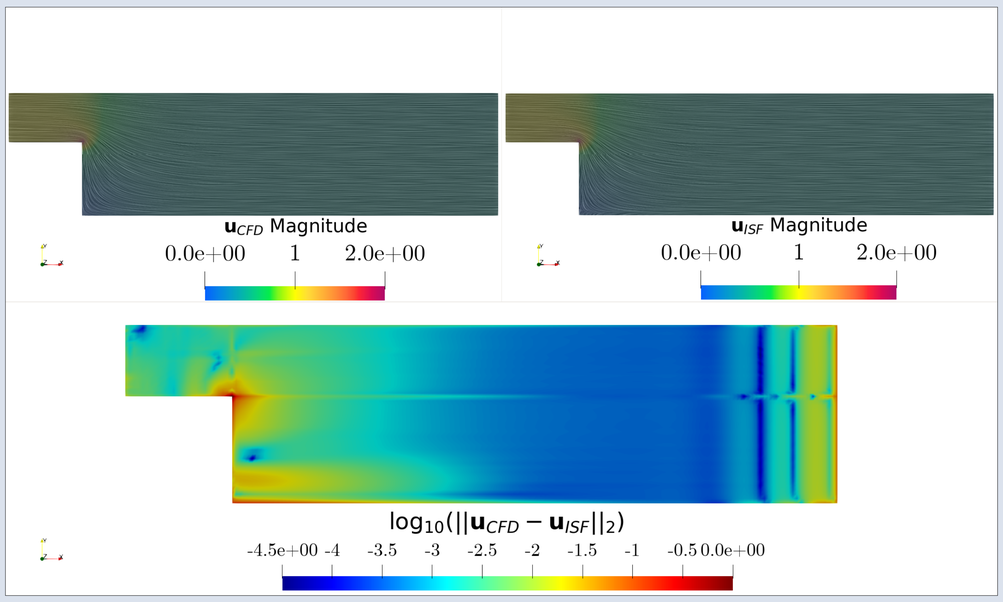

Figure 4. Comparison at $T = 10\,s$ between the ISF velocity streamlines and the **potential formulation** using OpenFOAM ([5]), using LIC (Line Integral Convolution). Below the norm of the difference in a logarithmic scale.

Conclusions

This work aims at studying the *Incompressible Schrödinger Flow*, an innovative technique for simulating inviscid incompressible fluids governed by the Euler equations.

- The **simulation time of ISF is much less** than a direct numerical solution of the Euler equations, since the **time step size is larger** .
- Implementation in a **Finite Element environment** .
- Accurate prediction of **vortex dynamics** .
- Similarities with the **potential flow** .